

CHAPTER 5

ESTIMATION OF FEED COMPOSITION THROUGH ANN MODELING

Modeling of FCC unit becomes complex due to the presence of large number of hydrocarbons in the feed which undergo a variety of reactions to yield cracking products. Laboratory analysis of these components at their molecular level is not an easy task and therefore, it is not possible to account for each individual feed component and its reactions in any realistic modeling exercise. Use of kinetic lumps has extensively been made by various researchers to model FCC units with varying success. The earliest attempt had only two lumps, one representing the feed and the second component was the product (Weekman and Nace, 1968). Three (Weekman., 1969), four (Lee et al., 1989) and five (Bollas et al., 2007) lump models have successively been used with improved success. Clearly, larger the number of lumps used, closer we approach to the real system.

In all these models, feed is represented by a single lump of average composition and molecular weight. If instead, feed analysis is available in terms of hydrocarbon groups and their relative proportions, the reaction kinetics can be better accounted for (Nace and Weekman, 1971). 10 and 12 lumps (Nace and Weekman, 1971; Jacob et al., 1976; Ellis et al 1998; Kumar et al., 1995; Cerqueira et al 1997) have been successfully used to model/simulate industrial FCC units with better prediction capabilities. Of these lumps, 6 to 8 were used to characterize feed and intermediate product (light fuel oil 221-343 °C) and remaining four for the final products namely gasoline, LPG, dry gas, and coke. Typically for the above models, accuracy of model prediction depends on realistic FCC feed characterization in terms of paraffins, naphthenes, aromatic rings and aromatic substituent groups. However, it

is not practical to undertake quantitative feed analysis in terms of hydrocarbon types before embarking on simulation every time particularly in a field laboratory, which is not usually equipped for such analysis.

The present work aims to develop an Artificial Neural Network based model which can use routinely measured properties of FCC feed such as density, ASTM distillation temperatures, Conradson carbon residue (CCR), sulfur and nitrogen content as inputs and provide detailed composition (wt percent of paraffins, naphthenes and aromatics) as output. Artificial neural network (ANN) type of modeling is most suited for this work since we do not know any functional relationship, even if it exists, between available inputs and desired outputs. Being a black box approach, ANN does not require, nor attempt to develop, any mathematical relation, linear or non linear, between input and output and yet can effectively serve as a tool to estimate the detailed composition of the feed required for FCCU modeling.

5.1 ARTIFICIAL NEURAL NETWORK (ANN) MODEL DEVELOPMENT

Artificial neural networks have been successfully implemented in the chemical industry, especially in the areas of dynamic modeling. Bhat and McAvoy, (1990) used neural network for dynamic modeling of pH in a CSTR and compared the results with traditional ARMA modeling. They found ANN models to be more accurate than ARMA models. Lately an increasing trend is observed to model the steady state processes also using ANN approach. The height equivalent of theoretical plate (HETP) and pressure drop for columns with structured packing were predicted by neural network model and the results were found to be more accurate than traditional semi-empirical model (Pollock and Eldridge, 2000; Whaley et al., 1999). Neural network models have also been developed for the prediction of heavy gas oil cracking

products (Sheikhattar and Zahedi, 2011) in hydrocracking (Elkamel et al., 1999), fluid catalytic cracking and catalytic reforming units (Zahedi et al., 2008). An ANN model was compared with a non-linear statistical model for FCC (Michalopoulos et al., 2001). The results showed the superiority of the ANN model in terms of prediction accuracy. An ANN model based optimizer to separate gas flow coming out of a hydrocracking reactor was developed (Altissimi et al., 1998) and the ANN results were compared with those obtained from the first principle models developed by Bayley et al, (1993). The ANN model was trained with data obtained from a commercial simulator (ASPEN PLUS) in absence of availability of sufficient plant data.

Neural networks have been used as soft sensors for the estimation of various parameters of crude petroleum column and prediction of product properties. (Ganguly et al., 2002; Murty et al., 1995; Lee and Chen, 1993). Presently, several refineries use ANN models as soft sensors to estimate output stream quality online and use this information to control distillation columns. Its applications are also shown in the chemical plants for detecting gross errors in the faulty sensors (Gupta and Narasimhan, 1993).

Conventionally, the desired network architecture is arrived at by a constructive (Wang et al, 1994) or a destructive (Bhat and McAvoy, 1992) method. With the development of evolutionary techniques such as genetic algorithm, it has become possible to design the network architecture optimally and directly (Boozarjomehry and Svrcek, 2001; Dam and Saraf, 2006). In the present study, however, the conventional constructive method has been used to design the networks starting from a small network and expanding it by adding more layers and neurons in the hidden layers.

A feed forward back propagation type of modeling approach has been followed in the present work which is most suited for numerical NN modeling. The number of neurons for feed forward neural networks is approximately proportional to the number of learning data sets (Fujita, 1992 and 1998). All neurons in the input layer carry information to the neurons in the hidden layers after multiplying each with corresponding synaptic weights. Each layer has mapping function (somatic operation) to be suitably chosen. Figure 5.1 shows a typical neural unit (neuron) with linear synaptic operation. Normally, one or two hidden layers are sufficient to estimate any function (Haykin, 1999). The hidden layer may consist of one or more layers which may be connected in series or parallel or a combination with the output layer.

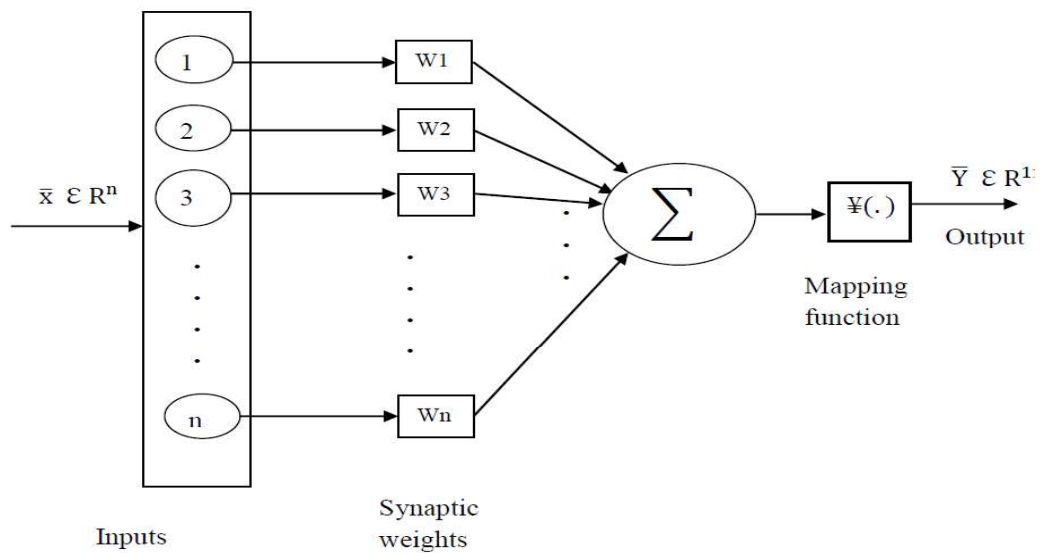


Figure 5.1: A typical neuron with linear synaptic operation.

In the present work, trial and error method was used to find near optimum ANN architecture for estimation of detailed feed composition (PNA). Neural Network Toolbox in MATLAB was used for the purpose. Various activation functions were investigated using different combinations for the hidden and the output layers. The numbers of neurons were also varied systematically in the hidden layers. The least

square error was optimized by addition of the neurons in the hidden layer. Finally, two hidden layers were found to be optimum.

In this algorithm, all information moves in the forward direction from input to output through hidden layers. Initial weights are arbitrarily assigned. During ANN training, the output is compared with the desired (experimental) known output and the error is back propagated to adjust the weights. Any non-linear optimization method whether local or global one can be used to optimize feed-forward neural network by changing the synaptic weights. The local searching optimization techniques are fundamentally limited to local solutions, while global ones attempt to avoid this limitation. The most popular optimization method has been Levenberg Marquardt algorithm based on variants of gradient, which was used in the present work. The training performance varies depending on the objective function and underlying error for a given problem and network configuration. After the network was fully trained and validated with new data sets, the network objective function was used to calculate the response to different inputs. Standard connections were used i.e network was fully connected between adjacent layers only. Figure 5.2 shows the ANN architecture used.

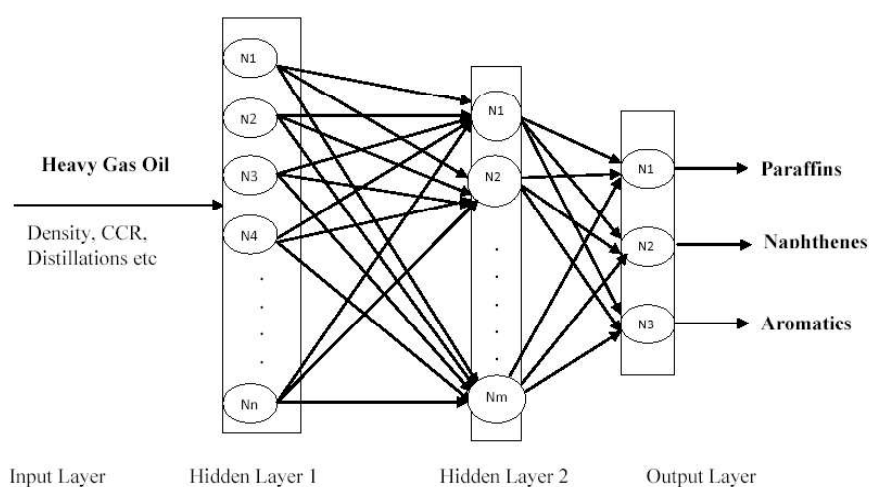


Figure 5.2: Neural network architecture with two hidden layers and an output layer.

5.2 MODEL PERFORMANCE EVALUATION CRITERIA

Following criteria were applied to evaluate the model performance

$$\text{Mean Square Error (MSE)} = \frac{1}{n} \sum_n (y_i^{\text{obs}} - y_i^{\text{pred}})^2 \quad (5.1)$$

$$\text{Root Mean Square Error (RMSE)} = \sqrt{(\text{MSE})} \quad (5.2)$$

$$\text{Percent deviation (PD)} = \frac{y_i^{\text{obs}} - y_i^{\text{pred}}}{y_i^{\text{obs}}} * 100 \quad (5.3)$$

Coefficient of determination (also called R²) as defined below, was also included here as it reflects the accuracy of prediction. For the perfect model R² = 1.

$$R^2 = \left(1 - \frac{\text{SSE}}{\text{SST}}\right) \quad (5.4)$$

Where SSE= sum of square error

$$\text{SSE} = \sum_n (y_i^{\text{obs}} - y_i^{\text{pred}})^2 \quad (5.5)$$

SST= total sum of square deviation from the mean

$$\text{SST} = \sum_n (y_i^{\text{obs}} - \bar{y}_i^{\text{obs}})^2 \quad (5.6)$$

A total of 28 samples were analyzed in the laboratory, of which 16 data sets covering the entire range of data (data range given in Table 5.1) were used for the development of ANN models (training). Another six data sets were used for testing the models and remaining 6 sets for model validation.

The developed ANN models are validated with remaining six data sets (input data range given in Table 5.2) representing FCC feeds with wide variation in compositions (paraffinic, naphthenic or aromatic in nature).

5.3 MODEL RESULTS

Two different ANN models (Model -1 and Model -2) have been developed in the present study. Model -1 predicts three output parameters: weight percents of

paraffin, naphthene and aromatic content of FCC feed from a single ANN architecture having three neurons in the output layer. Model -2 predicts the paraffin, naphthene and aromatic content individually from three different ANN architectures each with a single output neuron followed by normalization.

In ANN modeling there is always the big question about what should constitute the input parameters and there is no straight forward way to answer. One, therefore, tends to cautiously choose all possible inputs that are likely to influence the output. But this comes at a cost. Besides increasing computation load, particularly during training, it calls for larger data sets. Since all experimental data are prone to measurement errors, learning rates must be kept low resulting in further slowing down the training besides increasing model uncertainty and decreasing accuracy to convergence. It is therefore, desirable to use an optimal set of input parameters where the contribution of each input is more significant than the noise it adds.

Table 5.1: Range of Input Data sets Used for ANN Training

Parameters	Minimum	Maximum
Density, g/cc	0.8386	0.996
CCR (wt %)	0.04	1.61
Total Sulfur (wt %)	0.01	4.35
Total Nitrogen (ppm)	5	2089
SIMTBP (wt %)		
0 % ($^{\circ}$ C)	37	325
5% ($^{\circ}$ C)	224	409
10% ($^{\circ}$ C)	263	428
30% ($^{\circ}$ C)	337	458
50% ($^{\circ}$ C)	380	481
70% ($^{\circ}$ C)	407	506
90% ($^{\circ}$ C)	427	538
95% ($^{\circ}$ C)	436	552
100% ($^{\circ}$ C)	460	581
Paraffins (wt %)	4.9	27.7
Naphthenes (wt %)	15.4	55.9
Aromatics (wt %)	16.5	76.5

Tables 5.3 and 5.4 provide the details of ANN model architecture used in the present study along with activation functions for the 13 and 8 inputs respectively. In the present study, initially all the 13 measured properties were chosen as input, namely: density, ASTM distillation temperatures - IBP, 5% ,10 %,30 % ,50% ,70 % , 90 % , 95 % and FBP, Conradson carbon residue (CCR), total sulfur and total nitrogen. Subsequently sensitivity of each variable was examined. Based on the sensitivity study and intuitive reasoning, five of the 13 variables were dropped. IBP and FBP can be seldom determined with any amount of certainty while CCR, Sulfur and nitrogen content in VGO are unlikely to influence its PNA composition. Remaining 8 – variables were used as inputs. Results are presented for both 13 as well as 8 inputs.

Table 5.2: Range of Input Datasets Used for Model Validation

Parameters	Minimum	Maximum
Density, g/cc	0.8896	0.9335
CCR (wt %)	0.05	1.5
Total Sulfur (wt %)	0.0345	3.93
Total Nitrogen (ppm)	20	1240
SIMTBP (wt %)		
0 % ($^{\circ}\text{C}$)	199	304
5% ($^{\circ}\text{C}$)	285	389
10% ($^{\circ}\text{C}$)	313	413
30% ($^{\circ}\text{C}$)	370	455
50% ($^{\circ}\text{C}$)	410	481
70% ($^{\circ}\text{C}$)	446	506
90% ($^{\circ}\text{C}$)	491	535
95% ($^{\circ}\text{C}$)	510	544
100% ($^{\circ}\text{C}$)	538	581

Table 5.3: Summary of ANN Model Architectures for 13 Input Variables

Output Parameters	Model 1	Model 2		
	PNA	P	N	A
No. of input variables	13	13	13	13
No. of hidden layers	2	2	2	2
No. of neurons in layer-1	14	13	12	9
No. of neurons in layer-2	14	14	14	10
Activation function of layer-1	tansig	tansig	logsig	tansig
Activation function of layer-2	logsig	logsig	logsig	logsig
Activation function output layer	purelin	purelin	purelin	purelin
Performance function	mse	mse	msereg	msereg
Training function	trainlm	trainlm	trainlm	trainlm

Table 5.4: Summary of ANN Model Architectures for 8 Input Variables

Output Parameters	Model 1	Model 2		
	PNA	P	N	A
No. of input variables	8	8	8	8
No. of hidden layers	2	2	2	2
No. of neurons in layer-1	8	12	12	9
No. of neurons in layer-2	9	13	12	9
Transfer function of layer-1	logsig	tansig	tansig	tansig
Transfer function of layer-2	logsig	logsig	logsig	logsig
Transfer function output layer	purelin	purelin	purelin	purelin
Performance function	mse	mse	mse	msereg
Training function	trainlm	trainlm	trainlm	trainlm

Table 5.5 gives a comparison between experimental values and model predictions of gas oil composition for 13 inputs using both the models. Also included in this table are the percent deviations of the predicted values from the measured ones, root mean square error and R^2 values. Table 5.6 shows a similar comparison with 8 input variables. These comparisons were made for the validation sets of input data which means these data were not shown to the networks earlier.

Table 5.5: A Comparison of Model Predictions with Experimental Observations for Two Different ANN Models with 13 Input Variables (Validation Set): (a) Model 1 (b) Model 2

(a) Model 1: RMS error = 2.97, R^2 value = 0.98,

Sample ID	Paraffins			Naphthenes			Aromatics		
	Exp	Model 1	Dev %	Exp	Model 1	Dev %	Exp	Model 1	Dev %
1	15.8	14.6	7.71	35.9	36.7	-2.17	48.3	48.7	-0.91
2	14.4	15.1	-5.12	23.3	23.2	0.53	62.2	61.7	0.83
3	4.9	5.0	-1.56	21.1	22.2	-5.09	74.0	72.8	1.55
4	10.9	16.7	-52.74	29.9	26.9	9.90	59.2	56.4	4.71
5	10.4	9.2	11.40	18.5	24.6	-33.21	71.0	66.1	6.84
6	6.3	6.5	-3.96	26.2	30.6	-16.75	67.5	62.9	6.87

(b) Model2: RMS error = 2.04 ; R^2 value = 0.993

Sample ID	Paraffins			Naphthenes			Aromatics		
	Exp	Model 2	Dev %	Exp	Model 2	Dev %	Exp	Model 2	Dev %
1	15.8	16.9	-6.95	35.9	31.7	11.63	48.3	51.4	-6.37
2	14.4	12.6	12.50	23.3	26.9	-15.47	62.2	60.5	2.74
3	4.9	5.4	-9.15	21.1	23.8	-12.75	74.0	70.9	4.24
4	10.9	11.9	-9.75	29.9	29.5	1.19	59.2	58.5	1.20
5	10.4	12.1	-16.23	18.5	19.1	-3.23	71.0	68.8	3.08
6	6.3	6.4	-1.18	26.2	26.6	-1.47	67.5	67.0	0.68

While both the models have acceptable levels of error, Model 2 scores over Model 1 in all respects. This is in line with the common knowledge that it is better to develop architectures with only one neuron in the output layer i.e. each model should preferably predict one parameter. For Model 1, maximum deviation is 52.7 % whereas it is only 16.2 % for Model 2. Figure 5.3 shows a parity plot between predicted and measured percent compositions for Models 1 and 2 using 13 input parameters.

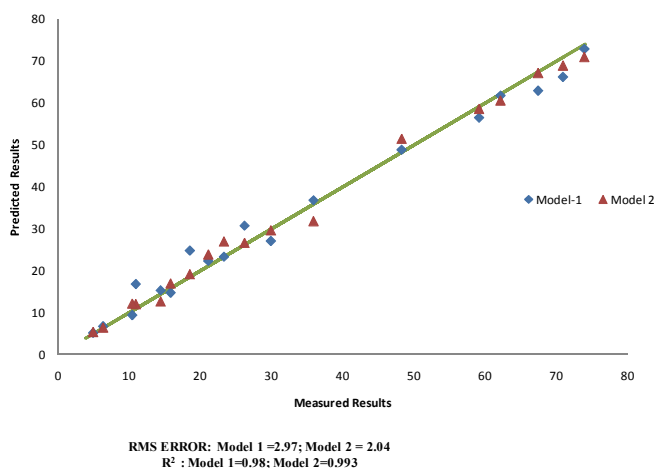


Figure 5.3: Parity plot between ANN predicted compositions and experimental values for models using 13 input variables.

Table 5.6 and Figure 5.4 provide similar information when only 8 inputs were used in place of 13. A comparison of Model 1 results for the two cases (with inputs 13 and 8) shows that fewer input parameters leads to less RMS error (2.09 against 2.97) and higher coefficient of determination, R^2 (0.993 against 0.98). Similar trend is observed for Model 2 also, however, less prominent.

Table 5.6: A Comparison of Model Predictions with Experimental Observations for Two Different ANN Models with 8 Input Variables (Validation Set)

Model 1: RMS error = 2.04; R2 value = 0.992,

Sample ID	Paraffins			Naphthenes			Aromatics		
	Exp	Model 1	Dev %	Exp	Model 1	Dev %	Exp	Model 1	Dev %
1	15.8	15.3	2.83	35.9	36.2	-0.71	48.3	48.5	-0.40
2	14.4	14.3	0.61	23.3	23.3	0.08	62.2	62.4	-0.33
3	4.9	4.8	2.31	21.1	20.8	1.20	74.0	74.4	-0.49
4	10.9	13.3	-22.03	29.9	26.2	12.40	59.2	60.5	-2.21
5	10.4	10.7	-3.27	18.5	23.3	-26.19	71.0	65.9	7.16
6	6.3	6.9	-9.92	26.2	24.6	6.17	67.5	68.5	-1.47

Model 2: RMS error = 1.71 ; R2 value = 0.994

Sample ID	Paraffins			Naphthenes			Aromatics		
	Exp	Model 2	Dev %	Exp	Model 2	Dev %	Exp	Model 2	Dev %
1	15.8	15.8	0.19	35.9	36.4	-1.47	48.3	47.8	1.03
2	14.4	13.5	6.25	23.3	22.6	2.91	62.2	63.9	-2.70
3	4.9	4.5	7.64	21.1	23.3	-10.46	74.0	72.2	2.48
4	10.9	13.4	-22.99	29.9	32.3	-8.08	59.2	54.3	8.31
5	10.4	9.4	9.74	18.5	18.9	-2.14	71.0	71.7	-1.01
6	6.3	6.8	-7.25	26.2	26.7	-1.93	67.5	66.5	1.43

Clearly the dropped input parameters namely IBP, FBP, total sulfur, total nitrogen and CCR do not have noticeable role in predicting the FCC feed compositions but their associated measurement errors affect the network model performance.

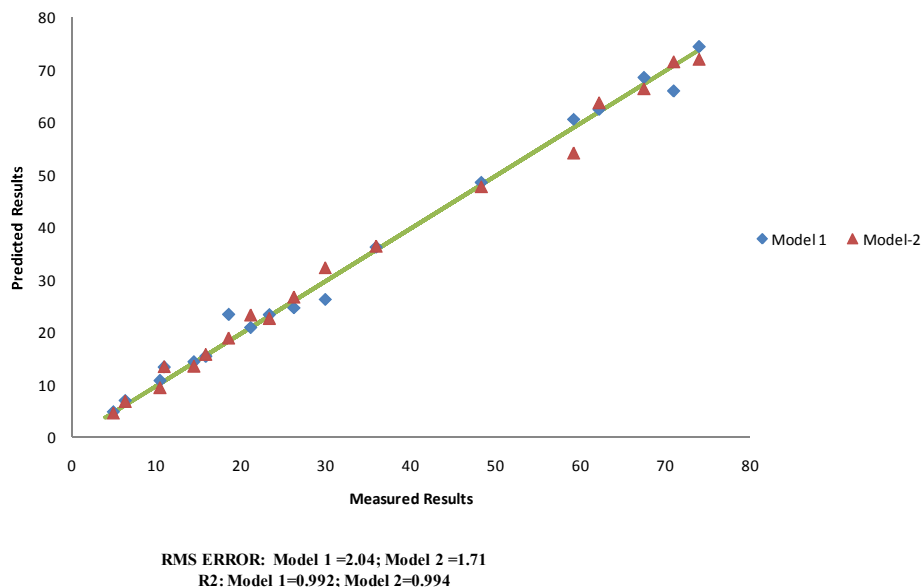


Figure 5.4: Parity plot between ANN predicted compositions and experimental values for models using 8 input variables.

This chapter presents an artificial neural network (ANN) model to predict detailed composition of FCC feed using routinely measured simple properties such as density, ASTM distillation temperatures, Conradson carbon residue content (CCR), sulfur and total nitrogen as inputs to the model. 60% of all the laboratory data sets were used to train the different networks, 20% for testing and remaining 20% were used for the validation. Several feed forward back propagation networks with different number of neurons in hidden layers were studied using Levenberg Marquardt (LM) training algorithm. Among different investigated models, the ANN model with 8 inputs, namely density and distillation temperatures except IBP, FBP to predict paraffin, naphthene and aromatic contents individually shows best agreement with the experimental results within permissible limit.